

Population Diversity and Comparative Demographic Dashboard of India

An interactive data visualization dashboard built on Census of India data

INTRODUCTION

India is one of the world's most diverse nations, with wide variation in population distribution, urbanization, household composition and geography across its states, union territories, districts and sub-districts. This project develops an interactive dashboard that turns official Census of India data into intuitive visual insights for policy-making, planning and research.

PROBLEM STATEMENT

Census data is available in large spreadsheets and statistical reports that are difficult for students, researchers, policymakers and the public to interpret. An interactive platform is needed to transform raw census data into meaningful visuals that enable quick comparison across regions.

OBJECTIVES

- Build an interactive dashboard using Census of India demographic data.
- Visualize population distribution across multiple administrative levels.
- Compare states, UTs, districts and sub-districts using key indicators.
- Present rural and urban population distributions through dynamic charts.
- Help users identify demographic patterns and regional disparities via interactive filters.

DATASET DESCRIPTION

Source dataset: "Number of Villages, Towns, Households, Population and Area," covering State/UT, District, Sub-District, Rural/Urban classification, villages (inhabited & uninhabited), towns, households, total/male/female population, area (sq. km) and population density — giving a comprehensive view of India's demographic landscape.

PROPOSED DASHBOARD FEATURES

- **KPI Cards** — KPI cards for total population, households, villages, towns and average density.
- **Interactive Map** — State-wise population distribution shown geographically.
- **Bar Charts** — Population, household, village and town comparisons by state.
- **Pie / Donut Charts** — Rural vs. urban and male vs. female population composition.
- **Treemap** — Population contribution of each state at a glance.
- **Scatter Plot** — Relationship between area and population density.
- **Ranking Tables** — Top 10 states by population, density and village count.
- **Filters** — Drill-down by State/UT, District, Sub-District, Rural/Urban, population range and density.

EXPECTED OUTCOMES

- Simplifies interpretation of complex census data for a general audience.
- Enables meaningful demographic comparisons across India.
- Supports academic research, classroom learning, and policy planning.
- Demonstrates practical application of BI and data visualization techniques.

CONCLUSION

The Population Diversity and Comparative Demographic Dashboard of India will transform official census statistics into an interactive decision-support tool — enabling users to explore India's demographic diversity efficiently while promoting evidence-based analysis for academic, administrative and policy-related purposes.

Link : <https://public.tableau.com/app/profile/waqar.ahmed.shaikh/viz/TowerOperationsCommandCenter/Operations>